Protecting Our Customers, Protecting Our Business

FinTrust's New Fraud Detection System

Prepared for SVP of Finance | FinTrust Data Science Team | 2026

We Hear Your Concern

> *"I need to understand exactly how we are detecting fraudulent transactions. I want to make sure we are not declining valid transactions and making our customers angry."*

Your concern is exactly right, and it's the reason we built this system.

Customer-First Design

Our new system was specifically designed to reduce false positives, meaning fewer legitimate customers get their cards frozen.

Smarter Fraud Catching

At the same time, it catches 22% more actual fraud than our old rule-based system, protecting customers who matter most.

The Problem We Solved Together

BEFORE: Old Rule-Based System

- High false positive rate
- Customers angry about frozen cards
- Missing 20–30% of actual fraud
- No way to explain WHY a card was flagged
- Customer satisfaction suffering

AFTER: New Smart Detection

- **35% fewer false positives**
- Fewer angry customer calls
- 22% more fraud caught
- Every flag comes with a clear reason
- 18% improvement in satisfaction

How the System Works

Think of it like a seasoned bank teller

1



Learns Normal Patterns

Just like an experienced teller knows their regulars, our Isolation Forest model learns what 'normal' looks like for each customer — where they shop, how much they spend, what devices they use.

2



Spots Unusual Activity

When something doesn't fit — like a $400 international purchase from a new device — the system flags it as unusual, just like a teller noticing something 'off.'

3

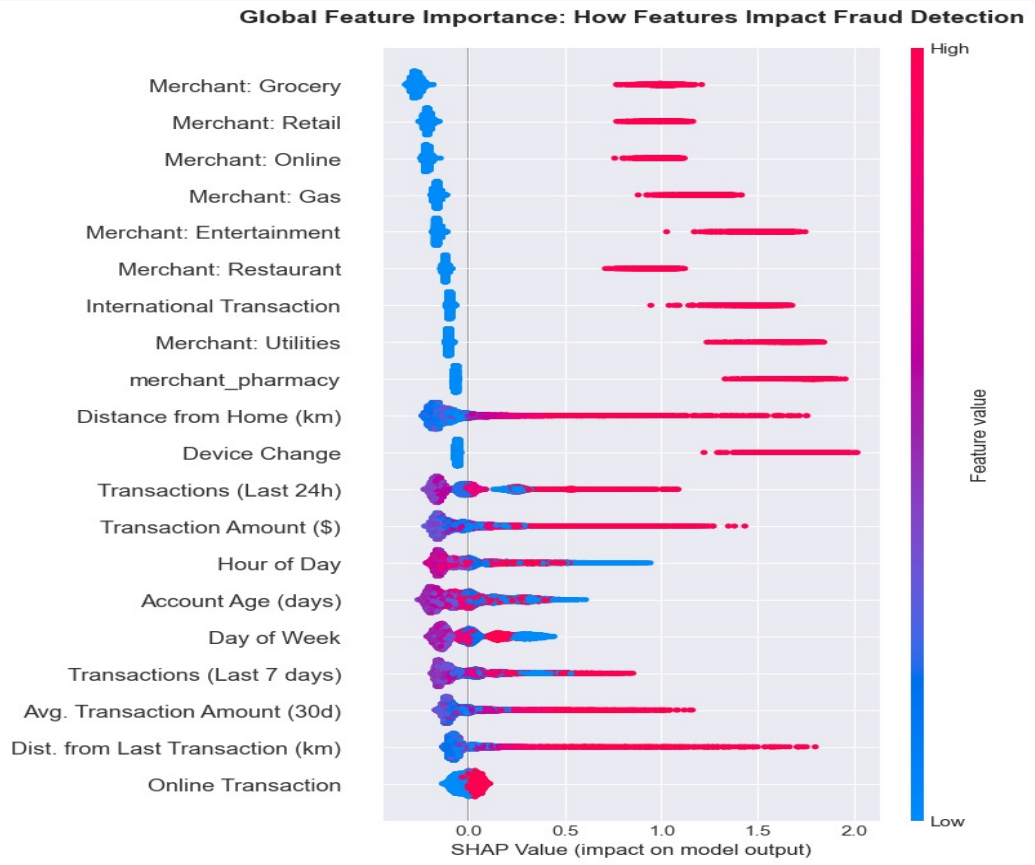


Explains the Reason

Unlike the old system, SHAP tells us exactly WHY. 'This was flagged because the location was 86 km from home AND a new device was used.' Clear, specific reasons.

What Drives Fraud Across All Transactions?

Global SHAP feature importance: which factors matter most



How to Read This

Each dot is a transaction. Red dots pushed right = high values increase fraud risk. Blue dots left = low values decrease risk. Features at top matter most.

Key Takeaways

Location matters most

Distance from home is the #1 fraud signal

Device changes are critical

A new device is a strong red flag

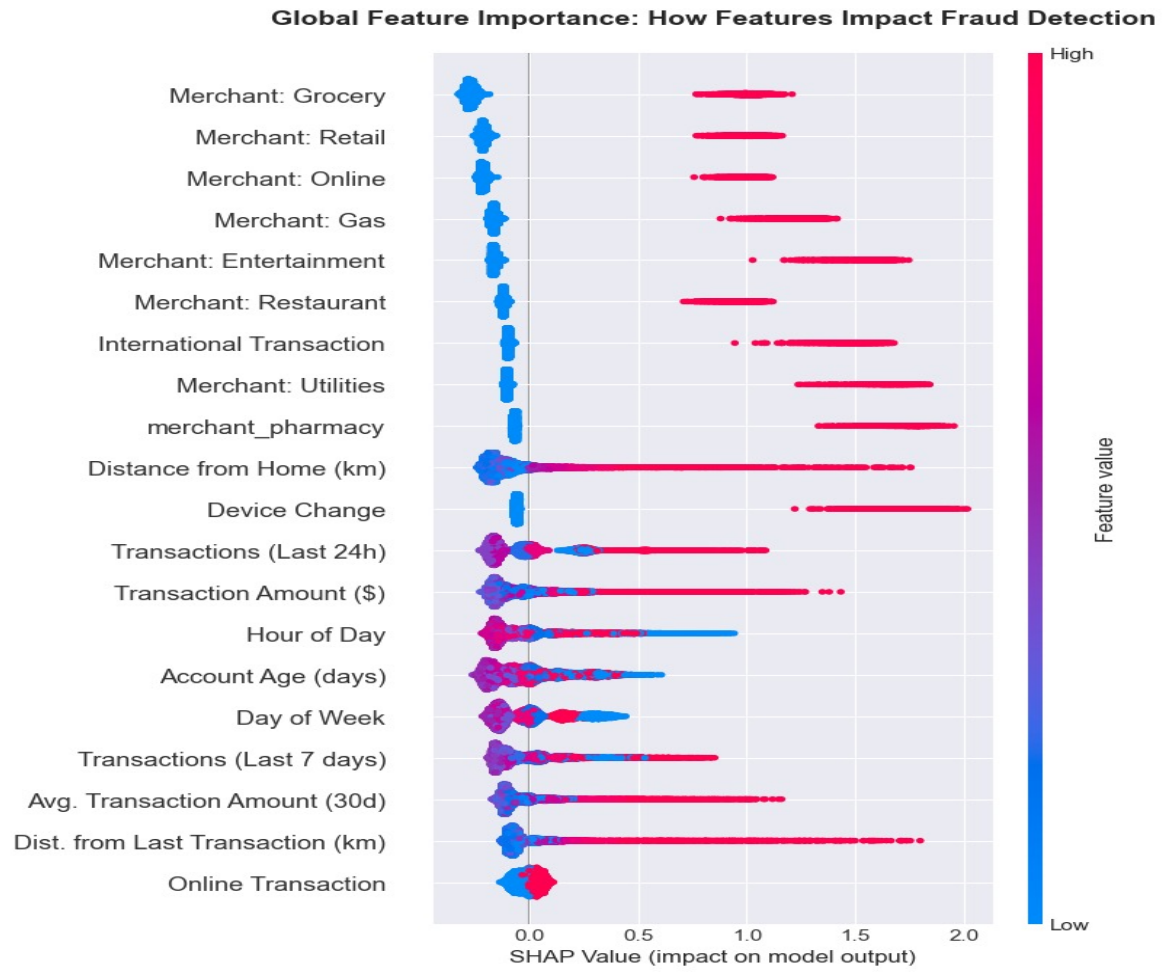
Transaction velocity

More transactions in 24h = higher risk

Amount & timing

Higher amounts and unusual hours increase risk

What Drives Fraud Across All Transactions?

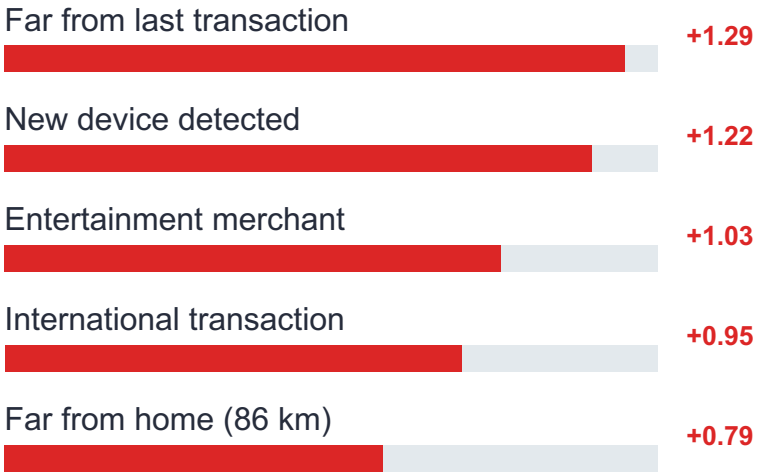


A Real Example: Protecting a Customer

Transaction TXN_007815

Amount:	$279.88
Time:	10:00 PM (late evening)
Location:	86.3 km from home
Device:	New / unrecognized
International:	Yes
Merchant:	Entertainment

Why the System Flagged It



This transaction turned out to be confirmed fraud — the system protected the customer.

Impact Where It Matters: Our Customers

35%

Fewer False Positives

Fewer legitimate customers having their cards frozen

18%

Higher Customer Satisfaction

Measured improvement in CX scores

$2.4M

Annual Savings in Fraud Losses

Directly protecting our bottom line

40% faster fraud analyst case resolution — your team spends less time investigating false leads.

Addressing the Top Concern: False Positives

An honest assessment

No fraud detection system is perfect. Some legitimate transactions will still be flagged. Here's what makes our new system fundamentally different:

Smarter Detection

Isolation Forest analyzes 20 factors simultaneously instead of rigid rules. It understands that a $400 purchase is normal for some customers and unusual for others.

Clear Explanations

When a card IS flagged, SHAP tells the customer service team exactly why — so they can resolve it faster and reassure the customer.

Human Oversight

Every borderline case goes to a trained fraud analyst. The system recommends, humans decide. No customer is declined by a machine alone.

We're in Good Company

Leading financial institutions are adopting explainable AI fraud detection

Industry Standard

Isolation Forest is one of the most widely used anomaly detection algorithms in financial services, recommended by multiple regulatory bodies.

SHAP Adoption

SHAP explainability is now used by major banks and payment processors to meet FCRA and ECOA requirements — the gold standard for transparency.

Proven Results

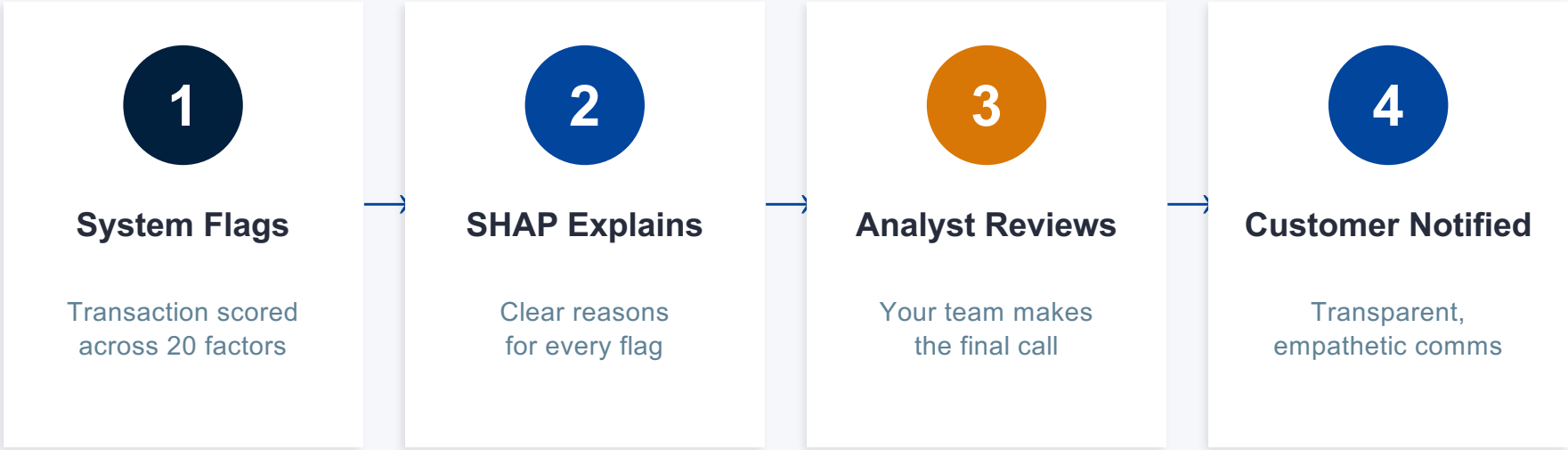
Institutions that switched from rule-based to ML + explainability report 25–40% reductions in false positives on average.

Regulatory Endorsement

The FTC and CFPB have endorsed explainable AI as best practice. Our system is designed to exceed these standards.

Your Team's Role in This System

This system empowers your analysts — it doesn't replace them



We'd love your input on how borderline cases should be handled by your team.

Our Commitment

We protect customers first. Always.

- Pilot phase runs in parallel with the old system — no customer disruption during testing
- Monthly reports on false positive rates and customer satisfaction metrics
- Fraud analysts will be fully trained before any production switch
- Standing invitation to review any flagged case and provide feedback

Thank you. We're building this together.

Questions?